

## **Abstract**

This is a test abstract to verify the functionality of the simplified English class file. This template is significantly simplified based on the original cumtthesis.cls file, retaining core LaTeX document structure elements, including document class declaration, chapter title formats, table of contents generation, main text environment, basic header and footer styles, and basic bibliography reference functionality.

The abstract environment test was successful, with font size and line spacing meeting expectations.

**Keywords:** LaTeX, template, simplified, thesis

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# **1 Introduction**

This is the content of the introduction chapter. This file demonstrates how to use the Chapter folder to organize the various chapters of the thesis.

## **1.1 Research Background**

This is the first section in the introduction, introducing the research background.

## **1.2 Research Significance**

This is the second section in the introduction, explaining the significance of the research.

## **1.3 Research Content**

This is the third section in the introduction, outlining the main research content.

## **2 Related Theories and Methods**

This is the content of Chapter 1, introducing related theories and methods.

### **2.1 Theoretical Foundation**

The first section of this chapter introduces the theoretical foundation.

#### **2.1.1 Basic Concepts**

This is a subsection about basic concepts.

#### **2.1.2 Related Research**

This is a subsection about related research.

### **2.2 Research Methods**

The second section of this chapter introduces research methods.

#### **2.2.1 Experimental Design**

This is a subsection about experimental design.

#### **2.2.2 Data Analysis**

This is a subsection about data analysis.

## 3 Function Testing

### 3.1 Mathematical Formula Testing

This is a mathematical formula test:

$$E = mc^2 \quad (3-1)$$

This is an inline formula test:  $a^2 + b^2 = c^2$

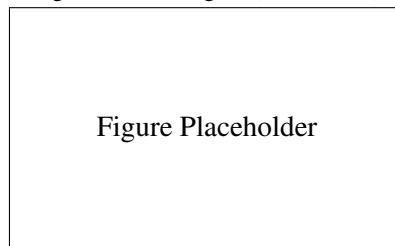
### 3.2 Table Testing

Table 3-1 Test Table

| Item   | Value | Unit |
|--------|-------|------|
| Length | 10    | cm   |
| Width  | 5     | cm   |
| Height | 2     | cm   |

### 3.3 Figure Testing

Figure 3-1 Test Figure (Placeholder)



### 3.4 Footnote Testing

This is a test sentence containing a footnote<sup>1</sup>. This is another test sentence containing a footnote<sup>2</sup>.

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<sup>1</sup>This is the first footnote

<sup>2</sup>This is the second footnote

## **Bibliography**

- [1] Jiang Xiao, Bai Lubin, Lou Xiayin, Li Mei, and Liu Hui. Research on tidal pattern extraction and demand forecasting of shared bicycles based on multi-scale spatiotemporal clustering. *Journal of Geoinformation Science*, 24(6):1047–1060, 2022.

## **Appendix 1**

This is the appendix content, which can include supplementary materials, data, etc.